SOLUTION REQUIREMENTS

Project Title

Rising Waters

Project Description

Rising Waters is an AI-powered Flood Prediction System developed using Machine Learning. The application predicts flood risk using rainfall and weather parameters, helping authorities issue early warnings and improve disaster preparedness.

Introduction

The purpose of this document is to describe the functional and non-functional requirements needed to successfully develop and deploy the Rising Waters application.

Functional Requirements

1. The system shall allow users to enter rainfall and weather parameters.

2. The system shall validate the entered input values.

3. The system shall process the input using the trained Machine Learning model.

4. The system shall predict flood risk as HIGH or LOW.

5. The system shall display the prediction result immediately.

6. The system shall provide a simple and user-friendly dashboard.

7. The system shall support multiple prediction requests.

Non-Functional Requirements

- High prediction accuracy.
- Fast response time.
- Easy-to-use interface.
- Reliable prediction results.
- Secure data processing.
- Cross-platform compatibility.
- Easy maintenance and future updates.

Software Requirements

- Python 3.x
- Streamlit
- Pandas
- NumPy
- Scikit-learn
- XGBoost
- Joblib

Hardware Requirements

• Intel Core i3 Processor or higher
• Minimum 4 GB RAM
• 10 GB Free Storage
• Internet Connection
• Windows, Ubuntu, or Linux Operating System

Expected Output

The application predicts flood risk accurately and displays the result in a professional dashboard. The system helps government authorities and disaster management teams take preventive actions before floods occur.

Benefits

• Early flood warning.
• Accurate prediction.
• Easy operation.
• Fast processing.
• Better disaster management.
• Improved public safety.

Conclusion

The Rising Waters system satisfies all essential software and hardware requirements needed for an efficient flood prediction application. The proposed solution is reliable, scalable, and suitable for real-world disaster management.